Lecture-23
Flow between plates continued, and the flow in a pipe

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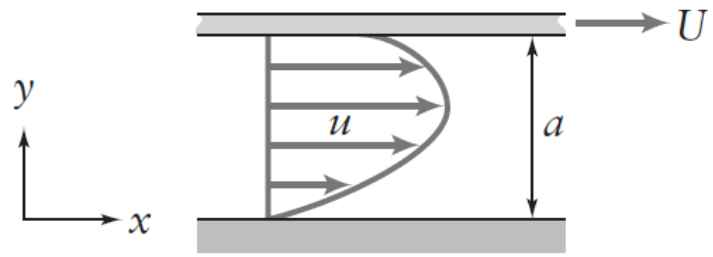
Learning Objectives

By the end of this lecture, you should be able to

1. Derive the velocity profile for **Couette-Poiseuille flow** between parallel plates.
2. Predict **flow reversal** in parallel plate flows by analyzing competing effects of **shear-** and **pressure-driven** mechanisms.
3. Solve for the **parabolic velocity distribution** for flow in a **circular pipe**
4. Calculate practical engineering parameters such as **volumetric flow rate**, **wall shear stress**, and **pressure drop**

Analytical Solution for Couette Poiseuille Flows

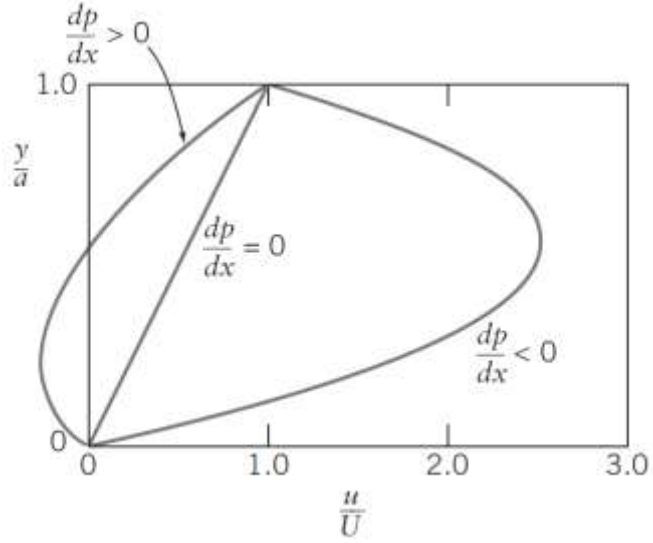
$$u = \frac{Uy}{a} + \frac{a^2}{2\mu}\left(\frac{dp}{dx}\right)\left[\left(\frac{y}{a}\right)^2 - \left(\frac{y}{a}\right)\right]$$



$$\tau_{yx} = \mu\frac{U}{a} + \frac{a^2}{2}\left(\frac{\partial p}{\partial x}\right)\left[\frac{2y}{a^2} - \frac{1}{a}\right] = \mu\frac{U}{a} + a\left(\frac{\partial p}{\partial x}\right)\left[\frac{y}{a} - \frac{1}{2}\right]$$

$$\frac{Q}{l} = \frac{Ua}{2} - \frac{1}{12\mu}\left(\frac{dp}{dx}\right)a^3$$

$$\overline{V} = \frac{Q}{A} = l\left[\frac{Ua}{2} - \frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right)a^3\right] \Big/ la = \frac{U}{2} - \frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right)a^2$$

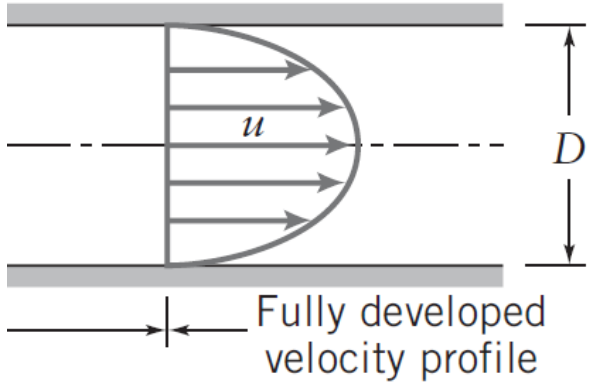


- When the pressure gradient is negative, the flow is always in the forward direction, but for **positive pressure gradients**, we may observe some **flow reversal**.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Fully developed laminar flow in a pipe

- Flow is **axisymmetric**, **steady**, and **fully developed**.
- We will use the Navier-Stokes equations in **cylindrical coordinates** to analyze this flow.



$$u = -\frac{R^2}{4\mu}\left(\frac{dp}{dx}\right)\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$\tau_{rx} = \mu\frac{du}{dr} = \frac{r}{2}\left(\frac{\partial p}{\partial x}\right)$$

$$Q = \int_A \vec{V}\cdot d\vec{A} = \int_0^R u 2\pi r\, dr = \int_0^R \frac{1}{4\mu}\left(\frac{\partial p}{\partial x}\right)(r^2 - R^2)2\pi r\, dr$$

$$Q = -\frac{\pi R^4}{8\mu}\left(\frac{\partial p}{\partial x}\right)$$

$$Q = -\frac{\pi R^4}{8\mu}\left[\frac{-\Delta p}{L}\right] = \frac{\pi \Delta p R^4}{8\mu L} = \frac{\pi \Delta p D^4}{128\mu L}$$

$$\overline{V} = \frac{Q}{A} = \frac{Q}{\pi R^2} = -\frac{R^2}{8\mu}\left(\frac{\partial p}{\partial x}\right)$$

$$u = u_{\max} = U = -\frac{R^2}{4\mu}\left(\frac{\partial p}{\partial x}\right) = 2\overline{V}$$

$$\frac{u}{U} = 1-\left(\frac{r}{R}\right)^2$$

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Revisiting the learning objectives

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